

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program – Sixth Semester – Fall 2024

Course Title: Artificial Intelligence

Course Code: D-326

Q.1: Answer the following short questions ($6 \times 5 = 30$)

Q1. Visual recognition is inherently challenging for a computer. Enlist at least three challenges.

Answer:

Visual recognition is difficult for computers due to several reasons:

1. **Variation in lighting:** Objects look different in bright or dim light.
2. **Different angles and views:** The same object can appear different from different positions.
3. **Background noise:** Irrelevant objects in the background confuse the system.
4. **Occlusion:** Sometimes objects are partially hidden.
5. **Object similarity:** Different objects may look very similar.
6. **Scale variation:** Objects may appear large or small depending on distance.
- 7.

Q2. Explain the concept of backpropagation in the context of training an ANN.

Answer:

Backpropagation is a learning algorithm used in Artificial Neural Networks:

1. It helps adjust weights to reduce error.
2. The network first performs a forward pass to calculate output.
3. Then it calculates error by comparing output with actual value.
4. The error is propagated backward through layers.
5. Weights are updated using gradient descent.
6. This process repeats until the error becomes minimal.

Q3. A city is trying to optimize traffic light timing using traffic density data. What factors would influence your fitness function?

Answer:

Important factors for the fitness function include:

1. **Traffic flow rate:** Number of vehicles passing through.
2. **Waiting time:** Time vehicles spend at signals.
3. **Queue length:** Number of cars waiting at intersections.
4. **Peak hour traffic:** Higher weight during busy times.
5. **Emergency vehicle priority:** Faster clearance when needed.
6. **Fuel consumption/emissions:** Reduce unnecessary stops.

Q4. What are the key components of First-Order Logic?

Answer:

First-Order Logic consists of the following components:

1. **Constants:** Represent specific objects (e.g., A, B).
2. **Variables:** Represent general objects (x, y).
3. **Predicates:** Describe relationships or properties.
4. **Functions:** Map objects to other objects.
5. **Quantifiers:** Universal ($\forall$) and Existential ($\exists$).
6. **Logical connectives:** AND, OR, NOT, IMPLIES.

Q5. What are the limitations of fuzzy logic and how can they be addressed in complex systems?

Answer:

Fuzzy logic has some limitations:

1. **No learning ability:** It cannot learn automatically.
2. **Rule complexity:** Too many rules make system complex.
3. **Subjectivity:** Membership functions depend on human judgment.
4. **Scaling issues:** Hard to apply in large systems.
5. **Accuracy issues:** May not give precise results.
6. **Solution:** Combine with neural networks (neuro-fuzzy systems) and optimization methods.

Q6. How is Natural Language Processing related to AI applications?

Answer:

Natural Language Processing (NLP) is an important part of AI:

1. It allows machines to understand human language.
2. Used in chatbots and virtual assistants.
3. Helps in translation between languages.
4. Enables sentiment analysis (positive/negative opinions).
5. Used in speech recognition systems.
6. Improves human-computer interaction

Q.2: Answer the following questions ($3 \times 10 = 30$)

Q.2(a) Suppose a genetic algorithm uses chromosomes of the form $x = a b c d e f g h$ with a fixed length of eight genes. Each gene can be any digit between 0 and 9. Let the fitness of an individual x be calculated as:

$$[f(x) = (a + b)^2 + (c + d) + (e + f) + (g + h)]$$

The initial population consists of four individuals with the following chromosomes:

- $x_1 = 3\ 2\ 9\ 2\ 0\ 2\ 8\ 5$
- $x_2 = 1\ 7\ 8\ 2\ 2\ 0\ 9\ 4$
- $x_3 = 5\ 6\ 4\ 0\ 3\ 5\ 3\ 2$
- $x_4 = 1\ 4\ 8\ 5\ 6\ 6\ 0\ 1$

Using the following information, perform the following operations:

1. Evaluate the fitness of each individual and rank them
2. Perform crossover operations (one-point and two-point crossover)
3. Perform mutation
4. Evaluate the new population
5. Find the maximum fitness

Answer:

Step 1: Fitness Evaluation

Using formula:

$$f(x) = (a+b)^2 + (c+d) + (e+f) + (g+h)$$

- **x1 = 32920285**
 $= (3+2)^2 + (9+2) + (0+2) + (8+5)$
 $= 25 + 11 + 2 + 13 = 51$
- **x2 = 17822094**
 $= (1+7)^2 + (8+2) + (2+0) + (9+4)$
 $= 64 + 10 + 2 + 13 = 89$
- **x3 = 56403532**
 $= (5+6)^2 + (4+0) + (3+5) + (3+2)$
 $= 121 + 4 + 8 + 5 = 138$
- **x4 = 14856601**
 $= (1+4)^2 + (8+5) + (6+6) + (0+1)$
 $= 25 + 13 + 12 + 1 = 51$

Step 2: Ranking (Highest to Lowest)

1. $x3 = 138$
2. $x2 = 89$
3. $x1 = 51$
4. $x4 = 51$

Step 3: Crossover

(i) One-Point Crossover (Middle Point)

Take x3 and x2:

x3 = 5640 | 3532

x2 = 1782 | 2094

Offspring:

- 56402094
- 17823532

(ii) Two-Point Crossover

Swap middle portion:

x3 = 56 | 40 35 | 32

x2 = 17 | 82 20 | 94

Offspring:

- 56822032
- 17403594

Step 4: Mutation

Randomly change one gene:

Example:

56402094 → **56412094** (0 → 1)

Step 5: Evaluate New Population

Apply same fitness formula to offspring to check improvement.

Step 6: Maximum Fitness

From initial population:

□ **Maximum fitness = 138 (x3)**

Q.2(b) Define Artificial Intelligence. Discuss its features, advantages, disadvantages, and major applications in detail.

Answer:

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence such as learning, reasoning, and decision-making.

Features of AI:

1. **Learning Ability:** AI systems learn from data and improve over time
2. **Reasoning:** Ability to solve problems logically
3. **Automation:** Performs tasks without human intervention
4. **Accuracy:** Reduces human errors
5. **Adaptability:** Adjusts to new situations

Advantages:

1. Reduces human effort
2. Works 24/7 without fatigue
3. High accuracy and speed
4. Helps in dangerous tasks
5. Improves decision-making

Disadvantages:

1. High development cost
2. Lack of creativity
3. Job replacement issues
4. Dependence on machines
5. Limited emotional intelligence

Applications of AI:

1. **Healthcare:** Disease diagnosis and medical imaging
2. **Education:** Intelligent tutoring systems
3. **Finance:** Fraud detection and risk management
4. **Transportation:** Self-driving cars
5. **Customer Service:** Chatbots and virtual assistants
6. **Security:** Face recognition systems

Q.2(c)What is Natural Language Processing (NLP)? Explain its components, techniques, applications, and challenges in detail.

Answer:

Natural Language Processing (NLP) is a subfield of Artificial Intelligence that focuses on enabling computers to understand, interpret, and generate human language.

Components of NLP:

1. **Lexical Analysis:** Breaking text into words (tokens)
2. **Syntax Analysis:** Checking grammatical structure
3. **Semantic Analysis:** Understanding meaning of words
4. **Pragmatic Analysis:** Understanding context

Techniques Used in NLP:

1. Tokenization
2. Stemming and Lemmatization
3. Parsing
4. Machine Learning and Deep Learning models

Applications of NLP:

1. Chatbots and virtual assistants
2. Language translation (e.g., Google Translate)
3. Speech recognition
4. Sentiment analysis
5. Text summarization
6. Search engines

Challenges in NLP:

1. Ambiguity of words
2. Understanding context
3. Different languages and dialects
4. Sarcasm and emotions
5. Complex grammar rules